P2 Chapter 9: Differentiation

Implicit Differentiation

Implicit Differentiation

You're used to differentiating expressions where y is the subject, e.g. $y = x^2 + 3x$. The relationship between x and y is 'explicit' in the sense we can directly calculate y from x. But what about **implicit** relations, e.g:

$$x^2 + y^2 = 8x \quad or \quad \cos(x + y) = \sin y$$

MIND-BLOWING MOMENT

$$y = x^2$$

$$\frac{d}{dx} \qquad \frac{dy}{dx} = 2x \qquad \frac{d}{dx}$$

When seeing $y = x^2$ and differentiating, you probably think you're just differentiating the x^2. But in fact, you're differentiating **both** sides of the equation! (with respect to x)

y (by definition) differentiates to $\frac{dy}{dx}$

To differentiate implicitly you only need to know 2 things:

- Differentiate each side of the equation (using chain rule if necessary).
- Remember that y differentiated with respect to x is, by definition, $\frac{dy}{dx}$

Examples

$$\frac{d}{dx}(y^2) = \boldsymbol{2y\frac{dy}{dx}}$$

Use chain rule. "bla squared" differentiated is "2 bla" (i.e. $2y$). And the inner function (i.e. just y) differentiated is $\frac{dy}{dx}$. Multiply them together and hey presto.

In general, when differentiating a function of y, but with respect to x, slap a $\frac{dy}{dx}$ on the end. i.e.

$$\frac{d}{dx}\big(f(y)\big) = f'(y)\frac{dy}{dx}$$

$$\frac{d}{dx}(\sin y) = \boldsymbol{\cos y\frac{dy}{dx}}$$

$$\frac{d}{dx}(e^y) = \boldsymbol{e^y\frac{dy}{dx}}$$

$$\frac{d}{dx}(xy) = \boldsymbol{x\frac{dy}{dx} + y}$$

You might be wondering, "If ax differentiates to a (where a is a constant), why doesn't xy differentiate to y?"
It's because y is **not a constant**.

$$\frac{d}{dx}\left(e^{x^2y}\right) = \boldsymbol{\left(x^2\frac{dy}{dx} + 2xy\right)e^{x^2y}}$$

$$\frac{d}{dx}(\tan(x+y)) = \boldsymbol{\left(1 + \frac{dy}{dx}\right)\sec^2(x+y)}$$

$$\frac{d}{dx}(x^2 + \cos y) = \boldsymbol{2x - \sin y\frac{dy}{dx}}$$

Exam Note: $\frac{d}{dx}(xy)$ is incredibly common.

Meatier Examples

[Textbook] Find $\frac{dy}{dx}$ in terms of x and y where $x^3 + x + y^3 + 3y = 6$

?

[Textbook] Find the value of $\frac{dy}{dx}$ at the point $(1,1)$, where $e^{2x} \ln y = x + y - 2$

?

Meatier Examples

[Textbook] Find $\frac{dy}{dx}$ in terms of x and y where $x^3 + x + y^3 + 3y = 6$

$$3x^2 + 1 + 3y^2\frac{dy}{dx} + 3\frac{dy}{dx} = 0$$

Differentiate both sides of the equation with respect to x (**Common error**: Forgetting the 6 becomes 0)

$$3y^2\frac{dy}{dx} + 3\frac{dy}{dx} = -3x^2 - 1$$

Make $\frac{dy}{dx}$ the subject (in the manner you would do at GCSE)

$$\frac{dy}{dx}(3y^2 + 3) = -3x^2 - 1$$

$$\frac{dy}{dx} = \frac{-3x^2 - 1}{3y^2 + 3}$$

[Textbook] Find the value of $\frac{dy}{dx}$ at the point $(1,1)$, where $e^{2x}\ln y = x + y - 2$

$$2e^{2x}\ln y + e^{2x}\left(\ln y\frac{dy}{dx}\right) = 1 + \frac{dy}{dx}$$

Fro Tip: Substitute in sooner rather than later. (i.e. No need to make $\frac{dy}{dx}$ the subject first)

When $x = 1, y = 1$,

$$e^2\frac{dy}{dx} = 1 + \frac{dy}{dx}$$

$$(e^2 - 1)\frac{dy}{dx} = 1$$

$$\frac{dy}{dx} = \frac{1}{e^2 - 1}$$

Frobservation: In Year 1 differentiation, you only ever needed the x value to calculate the gradient a particular point. In Year 2 the gradient can depend on x and y.

Test Your Understanding

C4 Jan 2008 Q5

A curve is described by the equation

$$x^3 - 4y^2 = 12xy.$$

(*a*) Find the coordinates of the two points on the curve where $x = -8$. **(3)**

(*b*) Find the gradient of the curve at each of these points. **(6)**

a) ?

b) ?

C4 June 2014(R) Q3

Hint for (b): Solve simultaneously with original equation.

$$x^2 + y^2 + 10x + 2y - 4xy = 10$$

(*a*) Find $\frac{dy}{dx}$ in terms of x and y, fully simplifying your answer. **(5)**

(*b*) Find the values of y for which $\frac{dy}{dx} = 0$. **(5)**

(a) ?

(b) ?

Exam Note: This type of question was quite common.

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(*b*) Find the values of y for which $\frac{dy}{dx} = 0$. **(5)**

Hint for (b): Solve simultaneously with original equation.

a) $-512 - 4y^2 = -96y$

$y^2 - 24y + 128 = 0$

$(y - 16)(y - 8) = 0$

Gives point $(-8,16), (-8,8)$

b) Implicitly differentiating:

$$3x^2 - 8y\frac{dy}{dx} = 12x\frac{dy}{dx} + 12y$$

$\frac{dy}{dx}$ not needed, but it's $\frac{3x^2-12y}{12x+8y}$

If $x = -8, y = 16, \frac{dy}{dx} = 0$

If $x = -8, y = 8, \frac{dy}{dx} = -3$

	$x^2 + y^2 + 10x + 2y - 4xy = 10$		
(a)	$\left\{\frac{\cancel{dy}}{\cancel{dx}} \times\right\}$ $2x + 2y\frac{dy}{dx} + 10 + 2\frac{dy}{dx} - \left(4y + 4x\frac{dy}{dx}\right) = 0$	See notes	M1 A1 M1
	$2x + 10 - 4y + (2y + 2 - 4x)\frac{dy}{dx} = 0$	Dependent on the first M1 mark.	dM1
	$\frac{dy}{dx} = \frac{2x + 10 - 4y}{4x - 2y - 2}$		
	Simplifying gives $\frac{dy}{dx} = \frac{x + 5 - 2y}{2x - y - 1}$ $\left\{= \frac{-x - 5 + 2y}{-2x + y + 1}\right\}$		A1 cso oe
			[5]
(b)	$\left\{\frac{dy}{dx} = 0 \Rightarrow\right\}$ $x + 5 - 2y = 0$		M1
	So $x = 2y - 5$,		
	$(2y - 5)^2 + y^2 + 10(2y - 5) + 2y - 4(2y - 5)y = 10$		M1
	$4y^2 - 20y + 25 + y^2 + 20y - 50 + 2y - 8y^2 + 20y = 10$		
	gives $-3y^2 + 22y - 35 = 0$ or $3y^2 - 22y + 35 = 0$	$3y^2 - 22y + 35 \{= 0\}$ see notes	A1 oe
	$(3y - 7)(y - 5) = 0$ and $y = \ldots$	Method mark for solving a quadratic equation.	ddM1
	$y = \frac{7}{3}, 5$	$\{y =\} \frac{7}{3}, 5$	A1 cao
			[5]

Exam Note: This type of question was quite common.

Exercise 9.3

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Homework Exercise

1 By writing $u = y^n$, and using the chain rule, show that $\frac{d}{dx}(y^n) = ny^{n-1}\frac{dy}{dx}$

2 Use the product rule to show that $\frac{d}{dx}(xy) = x\frac{dy}{dx} + y$.

3 Find an expression in terms of x and y for $\frac{dy}{dx}$, given that:

a $x^2 + y^3 = 2$ **b** $x^2 + 5y^2 = 14$ **c** $x^2 + 6x - 8y + 5y^2 = 13$

d $y^3 + 3x^2y - 4x = 0$ **e** $3y^2 - 2y + 2xy = x^3$ **f** $x = \frac{2y}{x^2 - y}$

g $(x - y)^4 = x + y + 5$ **h** $e^x y = xe^y$ **i** $\sqrt{xy} + x + y^2 = 0$

4 Find the equation of the tangent to the curve with implicit equation $x^2 + 3xy^2 - y^3 = 9$ at the point (2, 1).

5 Find the equation of the normal to the curve with implicit equation $(x + y)^3 = x^2 + y$ at the point (1, 0).

6 Find the coordinates of the points of zero gradient on the curve with implicit equation $x^2 + 4y^2 - 6x - 16y + 21 = 0$.

Problem-solving

Find $\frac{dy}{dx}$ then set the numerator equal to 0 to find the x-coordinate at the points of 0 gradient. You need to find two corresponding y-coordinates.

7 A curve C is described by the equation

$$2x^2 + 3y^2 - x + 6xy + 5 = 0$$

Find an equation of the tangent to C at the point (1, −2), giving your answer in the form $ax + by + c = 0$, where a, b and c are integers. **(7 marks)**

Homework Exercise

8 A curve C has equation

$$3^x = y - 2xy$$

Find the exact value of $\frac{dy}{dx}$ at the point on C with coordinates $(2, -3)$. **(7 marks)**

9 Find the gradient of the curve with equation

$$\ln(y^2) = \tfrac{1}{2}x\ln(x-1), \quad x > 1, \quad y > 0$$

at the point on the curve where $x = 4$. Give your answer as an exact value. **(7 marks)**

10 A curve C satisfies $\sin x + \cos y = 0.5$, where $-\pi < x < \pi$ and $-\pi < y < \pi$.

a Find an expression for $\frac{dy}{dx}$ **(2 marks)**

b Find the coordinates of the stationary points on C. **(5 marks)**

11 The curve C has the equation $ye^{-3x} - 3x = y^2$.

a Find $\frac{dy}{dx}$ in terms of x and y. **(5 marks)**

b Show that the equation of the tangent to C at the origin, O, is $y = 3x$. **(4 marks)**

Challenge

The curve C has implicit equation $6x + y^2 + 2xy = x^2$.

a Show that there are no points on the curve such that $\frac{dy}{dx} = 0$.

b Find the coordinates of the two points on C such that $\frac{dx}{dy} = 0$.

Homework Answers

1 Letting $u = y^n$, $\frac{du}{dy} = ny^{n-1}$

$$\frac{d}{dx}(y^n) = \frac{du}{dx} = \frac{du}{dy} \times \frac{dy}{dx} = ny^{n-1}\frac{dy}{dx}$$

2 $\frac{d}{dx}(xy) = x\frac{d}{dx}(y) + \frac{d}{dx}(x)y = x\frac{dy}{dx} + 1 \times y = x\frac{dy}{dx} + y$

3 **a** $-\frac{2x}{3y^2}$ **b** $-\frac{x}{5y}$ **c** $\frac{-3-x}{5y-4}$

d $\frac{4-6xy}{3x^2+3y^2}$ **e** $\frac{3x^2-2y}{6y-2+2x}$ **f** $\frac{3x^2-y}{2+x}$

g $\frac{4(x-y)^3-1}{1+4(x-y)^3}$ **h** $\frac{e^x y - e^y}{xe^y - e^x}$ **i** $\frac{-2\sqrt{xy}-y}{4y\sqrt{xy}+x}$

4 $y = -\frac{7}{9}x + \frac{23}{9}$

5 $y = 2x - 2$

6 (3, 1) and (3, 3)

7 $3x + 2y + 1 = 0$

8 $2 - 3\ln 3$

9 $\frac{1}{4}(4 + 3\ln 3)$

10 **a** $\frac{\cos x}{\sin y}$ **b** $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$ and $\left(\frac{\pi}{2}, -\frac{2\pi}{3}\right)$

11 **a** $\frac{3 + 3ye^{-3x}}{e^{-3x} - 2y}$

b At O, $\frac{dy}{dx} = \frac{3+0}{e^0 - 0} = 3$

So the tangent is $y - 0 = 3(x - 0)$, or $y = 3x$.

Challenge

a $6 + 2y\frac{dy}{dx} + 2y + 2x\frac{dy}{dx} = 2x \Rightarrow \frac{dy}{dx} = \frac{x-y-3}{y+x}$

So $\frac{dy}{dx} = 0 \Leftrightarrow x - y = 3$

Substitute: $6x + (x-3)^2 + 2x(x-3) = x^2$

So $2x^2 - 6x + 9 = 0$

Discriminant $= -36$, so no real solutions to quadratic.

Therefore no points on C s.t. $\frac{dy}{dx} = 0$.

b (0, 0) and (3, −3)